Linear Algebra I-II
Theorems & Lemmas

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1 Vector Spaces

1.1 Vector Spaces and Linear Subspaces

Definition 1.1: Vector Space

A **vector space** over a field K is a set V, endowed with two operations

$$\begin{aligned} + : V \times V \to V, &\qquad (v_1, v_2) \mapsto v_1 + v_2, \\ \cdot : K \times V \to V, &\qquad (\alpha, v) \mapsto \alpha \cdot v \end{aligned}$$

s.t. the following conditions (axioms) hold:

(V1) $\forall v_1, v_2, v_3 \in V : v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3,$

(V2) $\exists$ an element $0 \in V : 0 + v = v \qquad \forall v \in V,$

(V3) $\forall v \in V \; \exists v' \in V : v + v' = 0,$

(V4) $\forall v_1, v_2 \in V : v_1 + v_2 = v_2 + v_1,$

(V5) $\forall a, b \in K \; \forall v \in V : a \cdot (b \cdot v) = (a \cdot b) \cdot v,$

(V6) $\forall v \in V : 1 \cdot v = v,$

(V7) $\forall a \in K \; \forall v_1, v_2 \in V : a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2,$

(V8) $\forall a, b \in K \; \forall v \in V : (a + b) \cdot v = a \cdot v + b \cdot v.$

Lemma 1.2: Properties of Vector Spaces

Let V be a vector space over a field K. Then:

(a) *The 0 vector from (V2) is unique. To avoid confusion (e.g. with $0 \in K$) we will sometimes denote it by 0_V.*

(b) *The element v' (corresponding to v) from (V3) is unique. We'll denote it by $-v$. We'll also write $w - v := w + (-v)$ for all $w \in V$,*

(c) $\forall v \in V : 0 \cdot v = 0,$

(d) $\forall a \in K : a \cdot 0 = 0,$

(e) $\forall v \in V : -1 \cdot v = -v,$

(f) $\forall v \in V : -(-v) = v,$

(g) *If $a \cdot v = 0$ for some $a \in K, v \in V$, then either $a = 0$ or $v = 0$,*

(h) *Associativity for $+$ holds also for some of n elements. $v_1 + \ldots + v_n$ makes sense no matter how we put the brackets $v_1 + (v_2 + (\ldots + (v_{n-1} + v_n)))$ or $(v_1 + v_2) + (\ldots)$ etc.*

Example: The Coordinate Space K^n

Let K be a field and $n \in \mathbb{N}$.

$$K^n = \{(a_1, \dots, a_n) \mid a_i \in K, 1 \leq i \leq n\}.$$

K^n is a vector space over K if we endow it with the following operations:

- Let $v = (a_1, \dots, a_n), w = (b_1, \dots, b_n) \in K^n$.
$$v + w = (a_1, \dots, a_n) + (b_1, \dots, b_n) := (a_1 + b_1, \dots, a_n + b_n),$$
- For $a \in K$
$$a \cdot v = a \cdot (a_1, \dots, a_n) := (a \cdot a_1, \dots, a \cdot a_n)$$
- $0 := (0, \dots, 0) \in K^n$

With these operations K^n becomes a vector space over K.

Definition 1.3: Linear Subspace

Let V be a vector space over K. A subset $W \subseteq V$ is called a **linear subspace** of V (or just subspace of V), if the following holds

(LSS1) $W \neq \emptyset$,

(LSS2) $\forall w_1, w_2 \in W : w_1 + w_2 \in W$,

(LSS3) $\forall a \in K, w \in W : a \cdot w \in W$.

Lemma 1.4: Linear Subspace is a Vector Space

Let V be a vector space over K and $W \subseteq V$ a subspace. Then W is a vector space on its own, when endowed with the operations from V.

Lemma 1.5: Criterion for a Linear Subspace

Let V be a vector space over K, and $W \subseteq V$ a subset. Then W is a linear subspace of V if and only if the following holds:

(1) $0_V \in W$,

(2) $\forall a_1, a_2 \in K, w_1, w_2 \in W : a_1 w_1 + a_2 w_2 \in W$.

Example: Spaces of functions

Fix a field K and a non-empty set S. Define

$$K^S = \{f : S \to K \mid f \text{ is a function}\}.$$

Define $+$ and $\cdot$ on K^S as follows. Let $f, g \in K^S$, $a \in K$. Define

$$(f+g) : S \to K, \qquad S \ni x \xmapsto{f+g} f(x) + g(x),$$
$$(a \cdot f) : S \to K, \qquad S \ni x \xmapsto{a \cdot f} a \cdot f(x).$$

Now Let $S = [0,1]$ and take $K = \mathbb{R}$. Define $C(S) := \{f \in \mathbb{R}^S \mid f \text{ is continuous on } [0,1]\} \subseteq \mathbb{R}^S$. Then $C(S) \subseteq \mathbb{R}^S$ is a linear subspace.

1.2 Span of Sets

Definition 1.6: Span

Let $S \subseteq V$ be a subset. We define

$$\mathrm{Sp}(S) := \bigcap_{W \in \mathcal{N}} W,$$

where

$$\mathcal{N} := \{W \mid W \text{ is a subspace of } V \text{ and } S \subseteq W\},$$

i.e., the collection of all subspaces $W \subseteq V$ that contain S. $\mathrm{Sp}(S)$ is called the span of S (in V).

Lemma 1.7: Properties of the Span

(1) $\mathrm{Sp}(S) \subseteq V$ *is a linear subspace.*

(2) If $W \subseteq V$ *is a linear subspace and* $S \subseteq W$*, then* $\mathrm{Sp}(S) \subseteq W$.

In other words, $\mathrm{Sp}(S)$ *is a linear subspace of* V*, and moreover among those subspaces that contain* S*,* $\mathrm{Sp}(S)$ *is the smallest one.*

Definition 1.8: Linear Combination

Let $n \in \mathbb{N}$, $a_1, \dots, a_n \in K$, $v_1, \dots, v_n \in V$. A **linear combination** of the vectors $v_1, \dots, v_n$ with coefficients $a_1, \dots, a_n$ is the vector

$$a_1 v_1 + \ldots + a_n v_n = \sum_{i=1}^{n} a_i v_i \in V.$$

It is important to note that in this course consider only finitely many elements in the sums of linear combinations.

Lemma 1.9: Alternate Definition of Span

Let $\emptyset \neq S \subseteq V$*. Then*

$$\mathrm{Sp}(S) = \{a_1 v_1 + \ldots + a_n v_n \mid n \in \mathbb{N}, a_i \in K, v_i \in S \quad \forall 1 \leq i \leq n\}.$$

Remark 1.10. *The set* S *can be finite or infinite. Regardless of that, each linear combination uses only finite numbers of elements from* S.

Let $n \in \mathbb{N}$, $v_1, \ldots, v_n \in V$. We'll write

$$\mathrm{Sp}(v_1, \ldots, v_n) := \mathrm{Sp}(\{v_1, \ldots, v_n\}).$$

Remark 1.11. *It obviously holds that*

$$\mathrm{Sp}(v_1, \ldots, v_n) = \left\{ \sum_{i=1}^{n} \alpha_i v_1 \,\middle|\, \alpha_i \in K \quad \forall 1 \leq i \leq n \right\}.$$

Remark 1.12. *It follows from the definition of span that*

$$\mathrm{Sp}(\emptyset) = \{0_V\}.$$

Definition 1.13: Generating Set

Let V be a vect. sp. over K and $S \subseteq V$. We say that S **generates** (or **spans**) V if $\mathrm{Sp}(S) = V$. S is called a generating set of V if $\mathrm{Sp}(S) = W$, where $W \subseteq V$ is a subspace of V. We say that S **generates** (or **spans**) W.

Definition 1.14: Finite Dimensional

A vector space V is called **finite-dimensional** if $\exists$ a **finite** set $S \subseteq V$ such that $\mathrm{Sp}(S) = V$. Otherwise V is called **infinite-dimensional**.

1.3 Bases of Vector Spaces

1.3.1 Linear Independence

Let V be a vector space over K.

Definition 1.15: Linear Independence

Let $v_1, \ldots, v_n$ be a list of n vectors in V. We say that $v_1, \ldots, v_n$ is **linearly independent** if the only linear combination of $v_1, \ldots, v_n$ that gives $0 \in V$ is the trivial one. In other words,

$$\forall a_1, \ldots, a_n \in K : a_1 v_1 + \ldots + a_n v_n \quad \Longrightarrow \quad a_1 = 0, \ldots a_n = 0.$$

By definition, we say that the empty list of vectors $\emptyset$ is linearly independent. We say that a list of vectors is linearly dependent if, the list is not linearly independent.

Remark 1.16. *(1) 0 can always be written as a linear combination of any finite list of vectors* $0v_1 + \ldots + 0v_n = 0$, *i.e., the trivial linear combination.*

(2) If the list $v_1, \ldots, v_n$ *has two vectors that are the same, say* $v_k = v_l$ *for* $1 \leq k < l \leq n$, *then* $v_1, \ldots, v_n$ *is **not** linearly independent. Indeed,* $v_k + (-1)v_l = 0$.

(3) The order of the vectors in the list $v_1, \ldots, v_n$ *has no effect on linear independence. So, if there are no repetitions in* $v_1, \ldots, v_n$ *we can talk about linear independence of the set* $\{v_1, \ldots, v_n\}$.

Definition 1.17: Linear Independence in the Context of Families

Let $\mathcal{F} = \{v_i\}_{i \in I}$ be a familiy of vectors in V, indexed by a set I. We say that $\mathcal{F}$ is linearly independent if $\forall n \in \mathbb{N}$ and any sequence of distinct indices $i_1, \ldots, i_n \in I$, the list of vectors $v_{i_1}, \ldots, v_{i_n}$ is linearly independent. Of course, if in $\mathcal{F}$ there are any repetitions of vectors, i.e., if $\exists i, j, i \neq j$, s.t. $v_i = v_j$, then $\mathcal{F}$ cannot be linearly independent.

Lemma 1.18: Subsets of Linearly Independent Sets are Linearly Independent

If $\emptyset \neq S \subseteq V$ is linearly independent. Then every non-empty subset $S' \subseteq S$ is also linearly independent.

Definition 1.19: Linear Dependence

A family $\mathcal{F} = \{v_i\}_{i \in I}$ of vectors in V is called **linearly dependent** if it is not linearly independent.

1.3.2 Basis of a Vector Space

Definition 1.20: Basis of a Vector Space

A subset $S \subseteq V$ of a vector space is called a basis of V if the following holds:

(1) S is linearly independent.

(2) S generates (or spans) V, i.e., $\mathrm{Sp}(S) = V$.

Proposition 1.21: Equivalent Formulation of basis

A subset $S \subseteq V$ is a basis of V if and only if every $v \in V$ can be written in a unique way as a linear combination of vectors in S.

Proof. For simplicity we will assume that $|S| < \infty$. The case $|S| = \infty$ is similar.

$\underline{\Longleftarrow}$: We assume that every $v \in V$ can be written in a unique way as a linear combination of elements from S. Now, "can be written" means that $\mathrm{Sp}(S) = V$. Next write $S = \{v_1, \ldots, v_n\}$. Consider the vector $0 \in V$. Clearly

$$0 = 0 \cdot v_1 + \ldots + 0 \cdot v_n.$$

Since 0 can be written in a unique way as a linear combination of $v_1, \ldots, v_n$ it follows that

$$0 = a_1 v_1 + \ldots + a_n v_n \qquad \Longrightarrow \qquad a_1 = \ldots = a_n = 0.$$

Therefore, S is linearly independent, which proves that S is a basis of V.

$\underline{\Longrightarrow}$: We assume that S is a basis for V. By assumption $\mathrm{Sp}(S) = V$, so this means every $v \in V$ can be expressed as a linear combination of vectors from S. It remains to show that this unique.

Assume by contradiction that $\exists v \in V$ which has two different ways to be written as a linear combination of vectors from S, i.e.,

$$v = a_1 v_1 + \ldots + a_n v_n = b_1 v_1 + \ldots + b_n v_n$$

and $a_k \neq b_k$ for some $1 \leq k \leq n$. Then

$$0 = (a_1 - b_1)v_1 + \ldots + \underbrace{(a_k - b_k)}_{\neq 0} + \ldots + (a_n - b_n)v_n$$

and we deduce that 0 can be written as a non-trivial linear combination of elements from S. Therefore, S is linearly dependent, which is a contradiction to the fact that S is a basis of V. □

Example: Bases

$V = M_{m\times n}(K)$, $\{E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis, where

$$E_{ij} = \begin{pmatrix} 0 & \ldots & 0 & \ldots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \ldots & 1 & \ldots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \ldots & 0 & \ldots & 0 \end{pmatrix},$$

where the 1 is at the position (i, j) in the matrix.

Lemma 1.22: A

Let $v_1, \ldots, v_m \in V$ be a list of vectors which is linearly dependent. Then $\exists 1 \leq j \leq m$ s.t.:

(1) $v_j \in \mathrm{Sp}(v_1, \ldots, v_{j-1})$.

(2) $\mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_m) = \mathrm{Sp}(v_1, \ldots, v_m)$. *In other words, we can drop one of the vectors from the list without changing the span.*

Lemma 1.23: B

Same assumptions an in Lemma 1.22, but we assume in addition that $v_1, \ldots, v_k$ are linearly independent, where $1 \leq k \leq m$ is some index. Then $k < j$.

Lemma 1.24: C

Let $w_1, \ldots, w_n \in V$ s.t. $\mathrm{Sp}(w_1, \ldots, w_n) = V$, and let $v \in V$. Then $v, w_1, \ldots, w_n$ is linearly dependent.

Lemma 1.25: D

Let $v_1, \ldots, v_n \in V$ be s.t. $\mathrm{Sp}(v_1, \ldots, v_n) = V$, and let $u_1, \ldots, u_m \in V$ be linearly independent. Then $m \leq n$. Moreover, one can delete m vectors from the list $v_1, \ldots, v_n$ s.t. the remaining list, say $v_1', \ldots, v_n'$ together with $u_1, \ldots, u_m$ still span V, i.e., $\mathrm{Sp}(u_1, \ldots, u_m, v_1', \ldots, v_n') = V$.

Lemma 1.26: E

Let $w_1, \ldots, w_l \in V$ and assume that $w_j \notin \mathrm{Sp}(w_1, \ldots, w_{j-1})$ $\forall 1 \leq j \leq l$. Then the list $w_1, \ldots, w_l$ is linearly independent.

Lemma 1.27: F

Suppose that $\mathrm{Sp}(v_1, \ldots, v_n) = V$. *Then* $\exists$ *a subset of* $\{v_1, \ldots, v_n\}$ *which is a basis for* V.

Theorem 1.28: Bases Have the Same Number of Elements

Let V *be a finite dimensional vector space over* K. *Then* V *has a basis with finitely many elements. Moreover, every basis of* V *is finite and has the same number of elements.*

Definition 1.29: Dimension of a Vector Space

Let V be a finite dimensional vector space over K. We define the dimension of V to be the number $n \in \mathbb{N}$ of elements in a basis (any basis !) of V. We write $\dim V := n$. In case V is not finite dimensional we write $\dim V = \infty$.

Remark 1.30.

- $\dim V$ *is well defined as it does not depend on a specific choice of basis for* V.
- $\dim\{0\} = 0$, *b.c.* $\emptyset$ *is a basis for the space* $\{0\}$, *and* $|\emptyset| = 0$.

Summary

Let V be a finite dimensional vector space over K. Then:

(1) Every finite list of vectors that spans V contains a sublist which is a basis for V.

(2) Every linearly independent list can be extended to a basis of V.

Theorem 1.31: Equivalent Statements for Bases

Let V *be a vector space of* $\dim V = n \in \mathbb{N}$. *Let* $v_1, \ldots, v_n \in V$. *Then the following statements are equivalent:*

(1) $v_1, \ldots, v_n$ *are linearly independent.*

(2) $v_1, \ldots, v_n$ *generate* V, *i.e.,* $\mathrm{Sp}(v_1, \ldots, v_n) = V$.

(3) $v_1, \ldots, v_n$ *is a basis for* V.

Theorem 1.32: Consequences of Bases

Suppose that $n = \dim V < \infty$. *Let* $v_1, \ldots, v_k \in V$. *Then:*

(1) If $k < n$, *the list* $v_1, \ldots, v_k$ ***cannot*** *span* V.

(2) If $k > n$, *the list* $v_1, \ldots, v_k$ ***cannot*** *be linearly independent.*

Proposition 1.33: Dimension of Subspaces

Let V *be a finite dimensional vector space over* K. *Then every subspace* $U \subseteq V$ *is also finite dimensional. Moreover,* $\dim U \leq \dim V$ *with equality if and only if* $U = V$.

1.4 Row and Column Spaces

Definition 1.34: L

t $A \in M_{m\times n}(K)$ and $u_1, \ldots, u_m \in K^n$ the rows of A and $v_1, \ldots, v_n \in K^m$ the columns of A, i.e.,

$$A = \begin{pmatrix} — & u_1 & — \\ & \vdots & \\ — & u_m & — \end{pmatrix} = \begin{pmatrix} | & & | \\ v_1 & \ldots & v_n \\ | & & | \end{pmatrix}$$